http://leap.ee.iisc.ac.in/sriram/teaching/ADL2020/

E9: 309 ADL 21-12-2020

Housekeeping

- Midterm project II presentations
  - Done during Dec. 29,30th
  - Same format as previous evaluation
- Midterm project III
  - Abstract submission deadline (Jan 10th)
    - ✓ Evaluation after final exam (1st week of Feb) → Exact dates announced later
- Final Exam (as per IISc schedule)
  - ✓ Jan 22nd afternoon! → Format will be similar to mid-term.

Sat. Jan 23 Afternoon

Recap from previous lectures

- Analyzing trained neural networks
  - ✓ Hierachical representations ✓
  - ✓ Activation maps to determine saliency ✓
  - ✓ Incorporating attention mechanism for improved explainability ✓

relevance weighting

Today's lecture

✴️ Advancing deep learning models explainability

- ✓ Causal inference and structural causal models
- ✓ Pruning based analysis of deep learning
- ✓ Adversarial attacks

Causal inference

X smoke $\rightarrow$ Y lung cancer $= 0$ $= 1$

- Causal inference
  - Deriving the causal connection between conditions that cause an effect.
- Three levels of causation
  - association Seeing and observing the environment. Is the incidence of lung cancer higher among smokers?

    X Y
  - intervention Doing and intervening in the environment. How do we reduce lung cancer? What is the effect if we ban cigarettes?
  - counterfactuals Imagining, restrospection, understanding the environment. What if I had not smoked for the last two years?

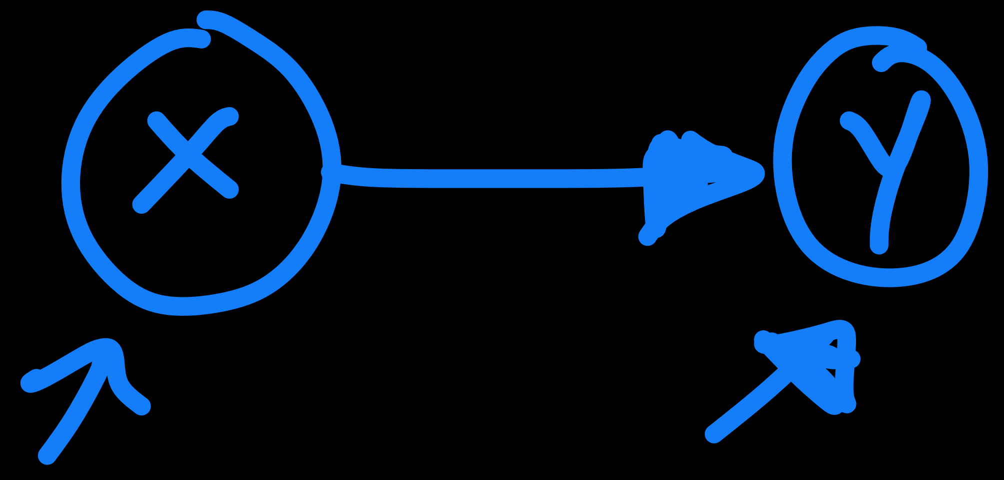
Correlations do not mean causations

Causality

- Causality can be modeled using random variables and conditioning

- On an intuitive level, the idea is that the factorization of the joint distribution P(Cause, Effect) into P(Cause)*P(Effect | Cause) typically yields models of lower total complexity than the factorization into P(Effect)*P(Cause | Effect).

- Causal inference can be described using directed graphs

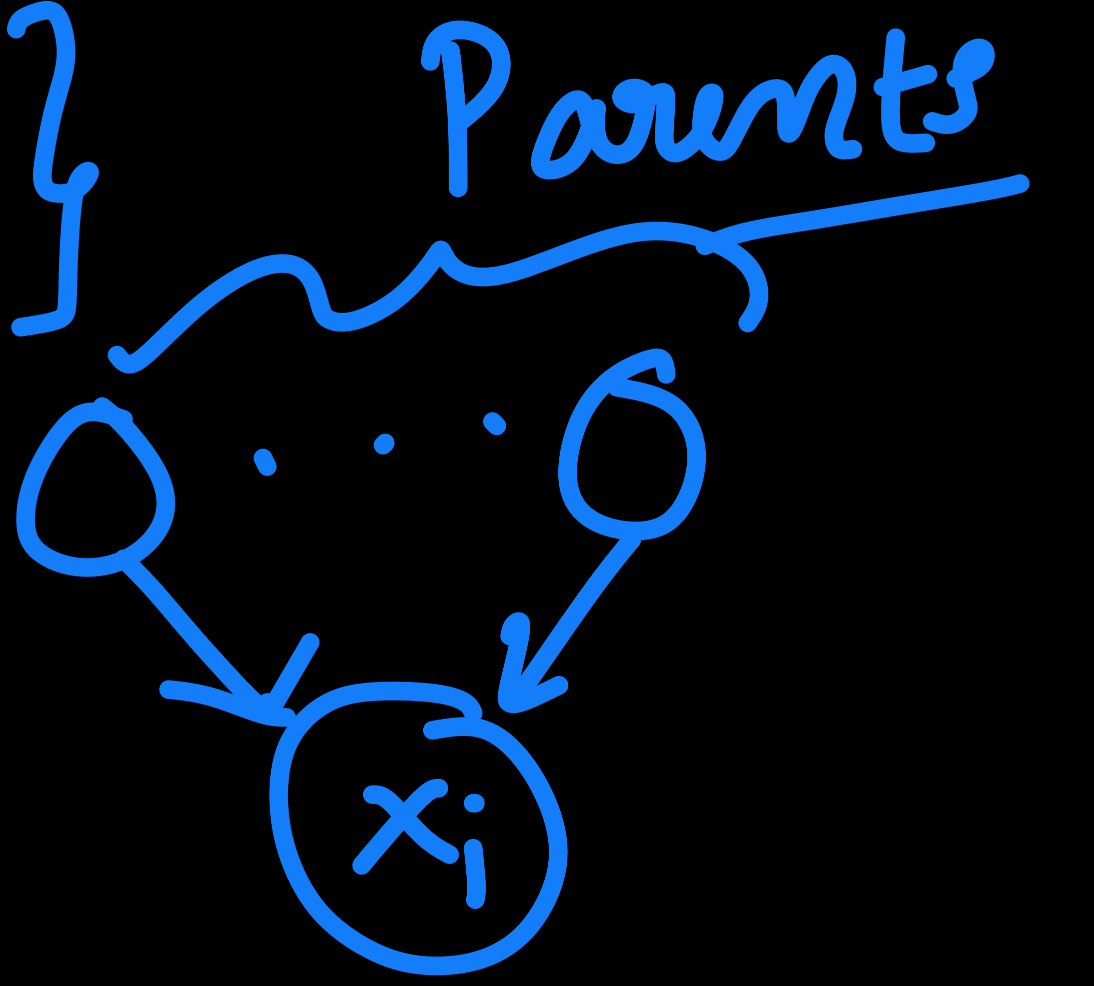


System $X = \{x_1, \dots \; x_N\}$

$$p(x_1 \dots x_N) = \prod_{i=1}^{N} p(x_i \mid Pa(x_i))$$

$$P_a(x_i) \subset \{X - x_i\}$$

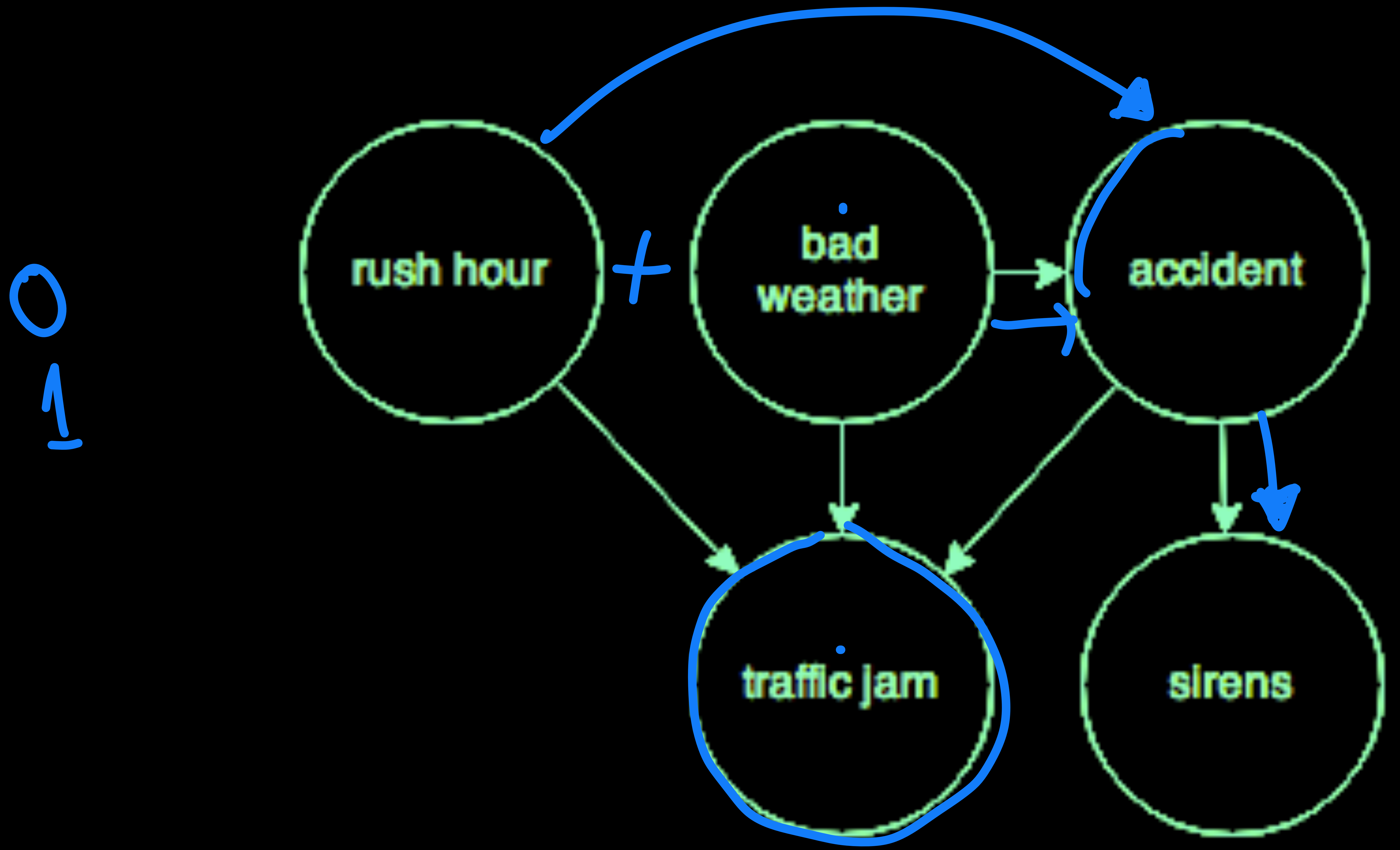
$\hookrightarrow$ parent nodes



$x_i \rightarrow$ child node

- Vertices - random variables
- Edges - cond. prob.

Causal modeling - example

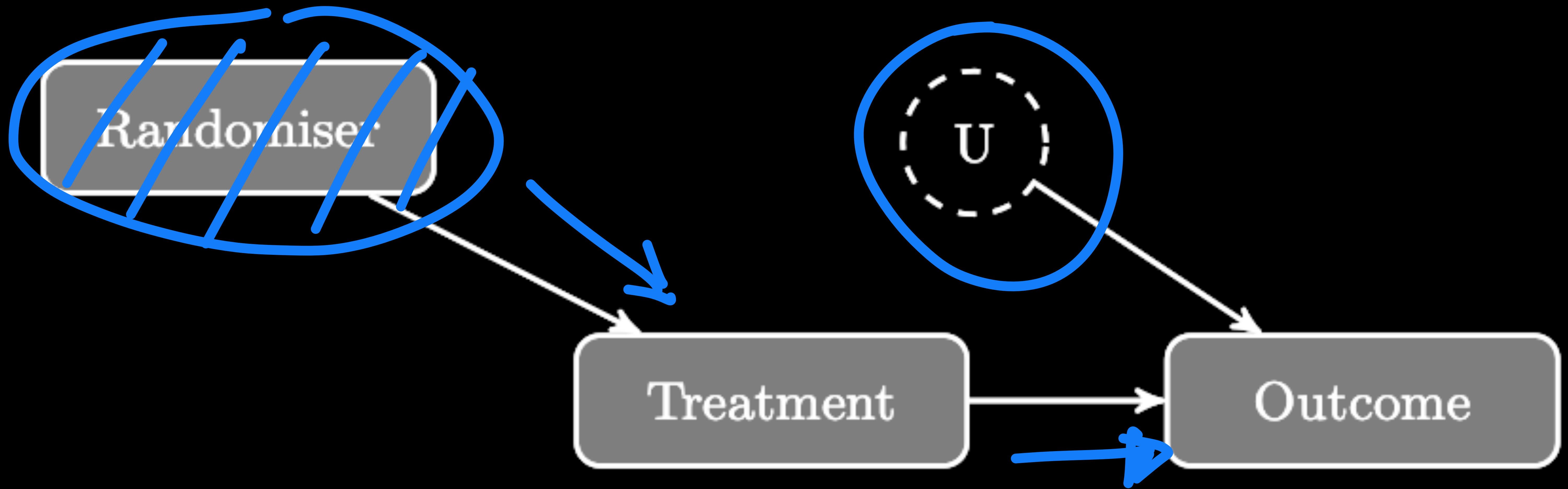


Principle

Reichenbachs common cause principle

- ✓ If two random variables X and Y are statistically dependent ($X \not\perp\!\!\!\perp Y$), then there exists a third variable Z that causally influences both. As a special case, Z may coincide with either X or Y . Furthermore, this variable Z screens X and Y from each other in the sense that given Z, they become independent, $X \perp\!\!\!\perp Y \mid Z$.

Causality in today's context

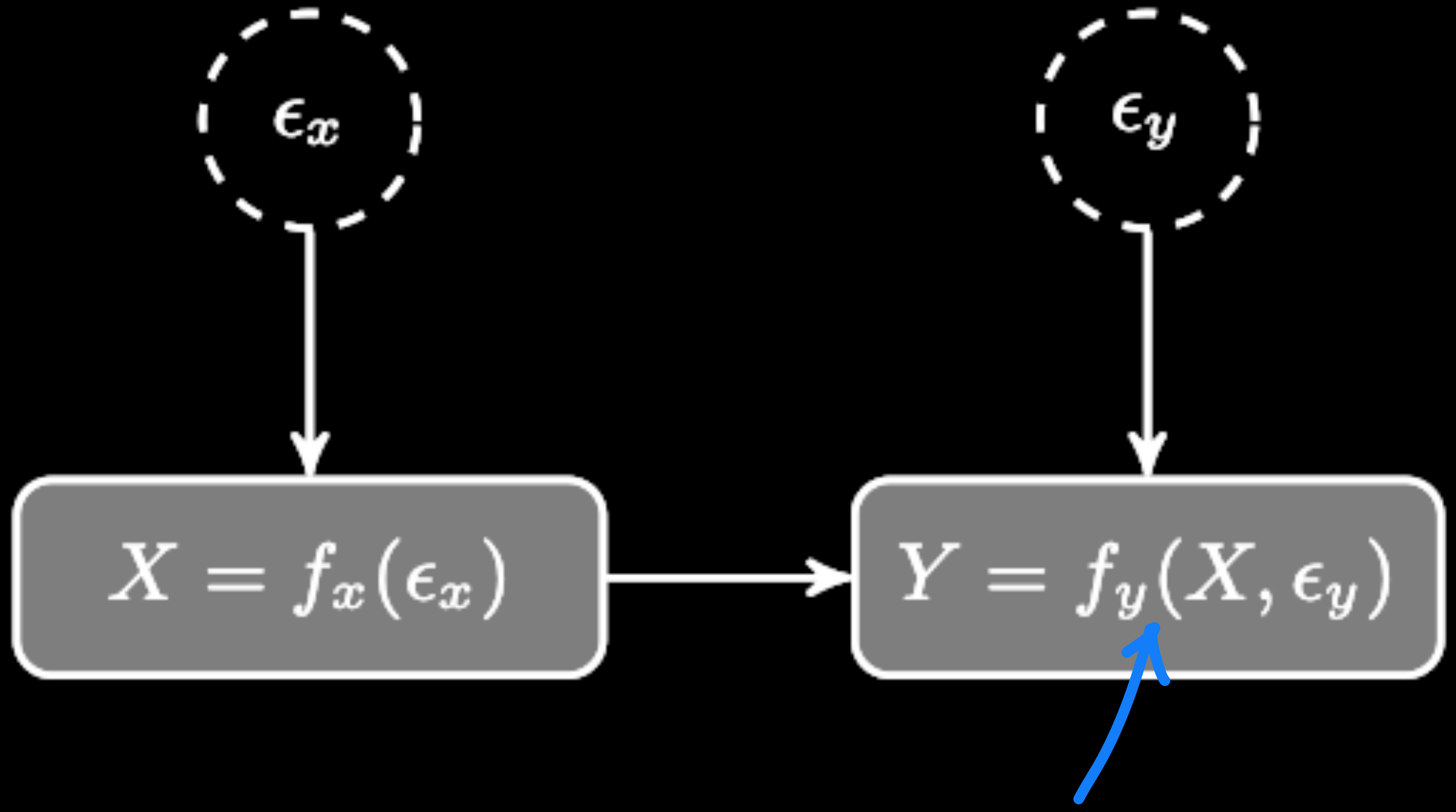


- Suppose a pharmaceutical company wants to assess the effectiveness of a new vaccine on recovery from a given illness. This is typically tested by taking a large group of representative patients and randomly assigning half of them to a treatment group (who receive the drug) and the other half to a control group (who receive a placebo). The goal is to determine the clinical impacts of the drug by comparing the differences between the outcomes for the two groups (in this case, simplified to only two outcomes - recovery or non-recovery). We will use the variable X (1 = drug, 0 = placebo) to represent the treatment each person receives and Y (1 = recover, 0 = not recover) to describe the outcome.

Structural Causal Equations

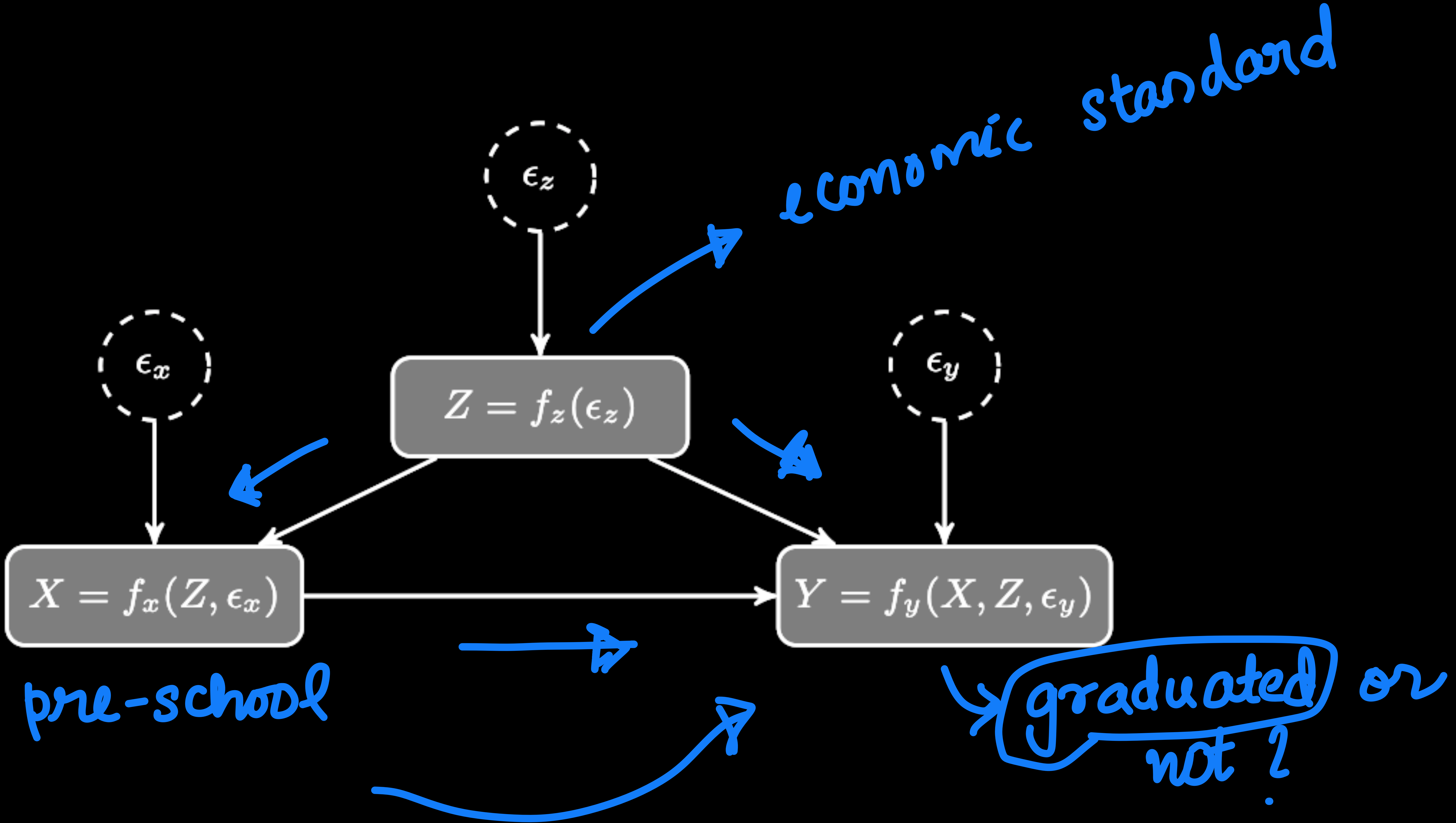
- Example of two variable causal model

$\epsilon_x \perp \epsilon_y$



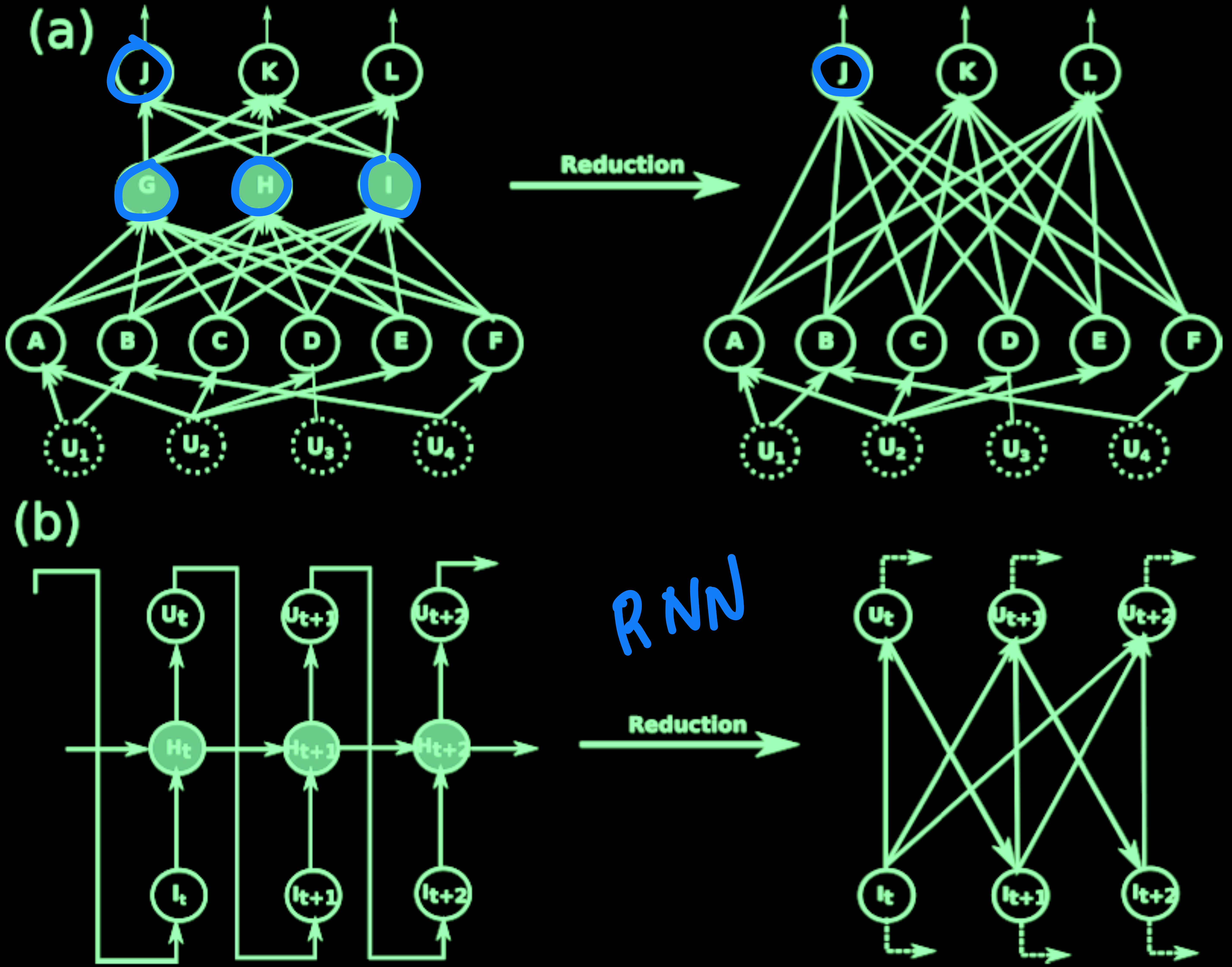
Structural Causal Equations

- Example of three variable causal model



Causal inference and deep learning

- Visualizing deep networks as directed graphs.
- Layers form dependency graphs
- Explicit computation of probabilities are often intractable
- Need for heuristics and approximations.



Pruning based approach to analyzing/compressing

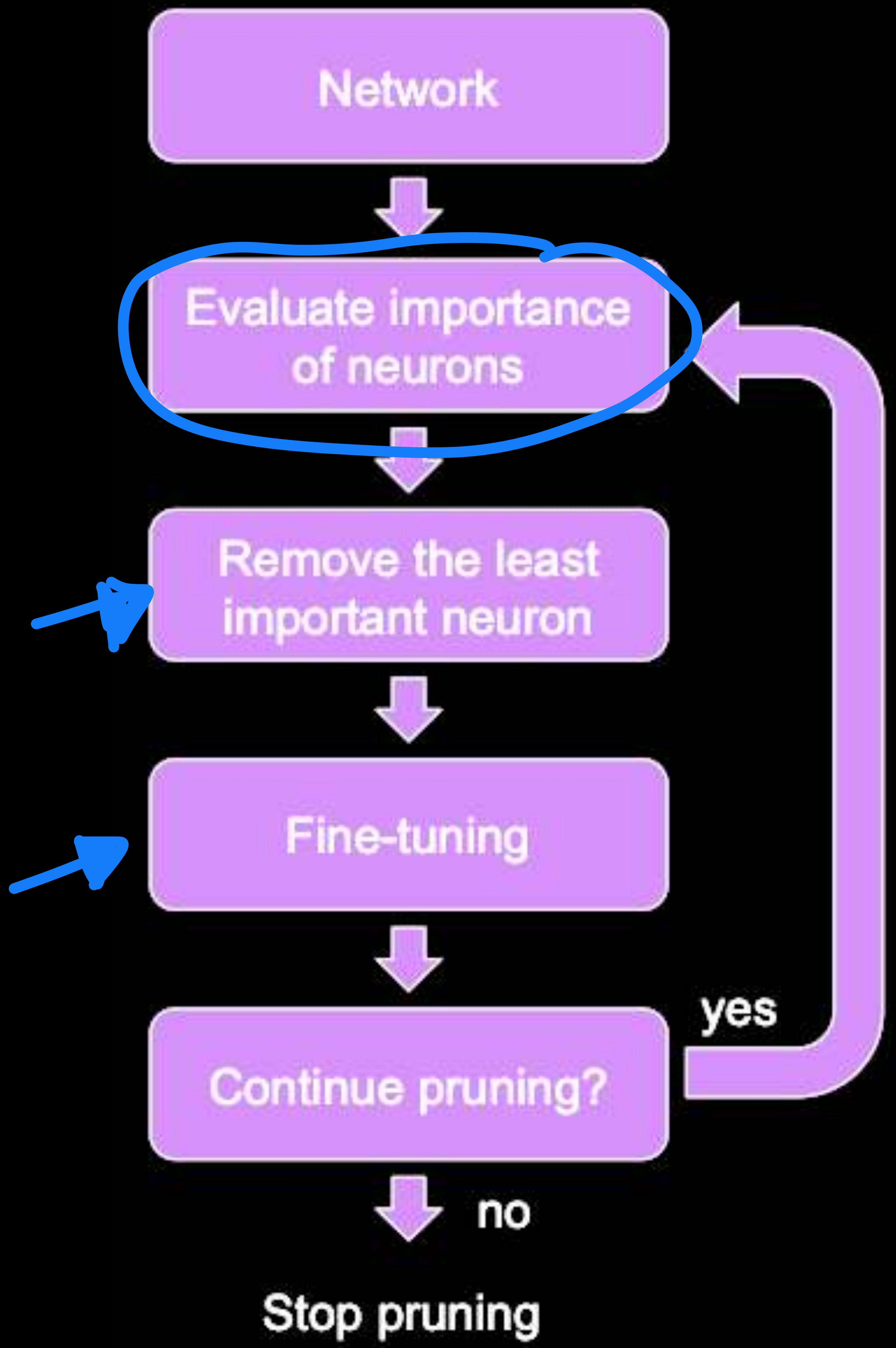
Published as a conference paper at ICLR 2017

PRUNING CONVOLUTIONAL NEURAL NETWORKS FOR RESOURCE EFFICIENT INFERENCE

Pavlo Molchanov, Stephen Tyree, Tero Karras, Timo Aila, Jan Kautz
NVIDIA
{pmolchanov, styree, tkarras, taila, jkautz}@nvidia.com

Pruning based approach to analyzing/compressing

- Removing connections of a learned neural network
  - Analyzing the effect of this intervention on the output of the model.
    - ✓ Example of intervention based causal model analysis.



Pruning based approach to analyzing/compressing

- Input data and targets

$$\overbrace{\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots \mathbf{x}_N\}}^{\text{data}} \quad \overbrace{\mathcal{Y} = \{\mathbf{y}_1, \mathbf{y}_2, \ldots \mathbf{y}_N\}}^{\text{labels}} \quad \overbrace{\mathcal{D} = \{\mathcal{X}, \mathcal{Y}\}}^{\text{Training}}$$

- Model parameters

$$\mathcal{W} = \{(\mathbf{W}^1, \mathbf{b}^1), \ldots, (\mathbf{W}^L, \mathbf{b}^L)\}$$

$W^1 = w^1_1 \ \ldots \ w^1_{k_1}$

- Error at the output

$$E(\mathcal{D}|\mathcal{W})$$

Pruning based approach to analyzing/compressing

- Problem formulation

$$\min_{\hat{W}} \left[\underbrace{|E(\mathcal{D}|W)} - \underbrace{E(\mathcal{D}|\hat{W})}| \right] \quad \text{such that} \quad \underbrace{||\hat{W}||_0 \leq B}$$

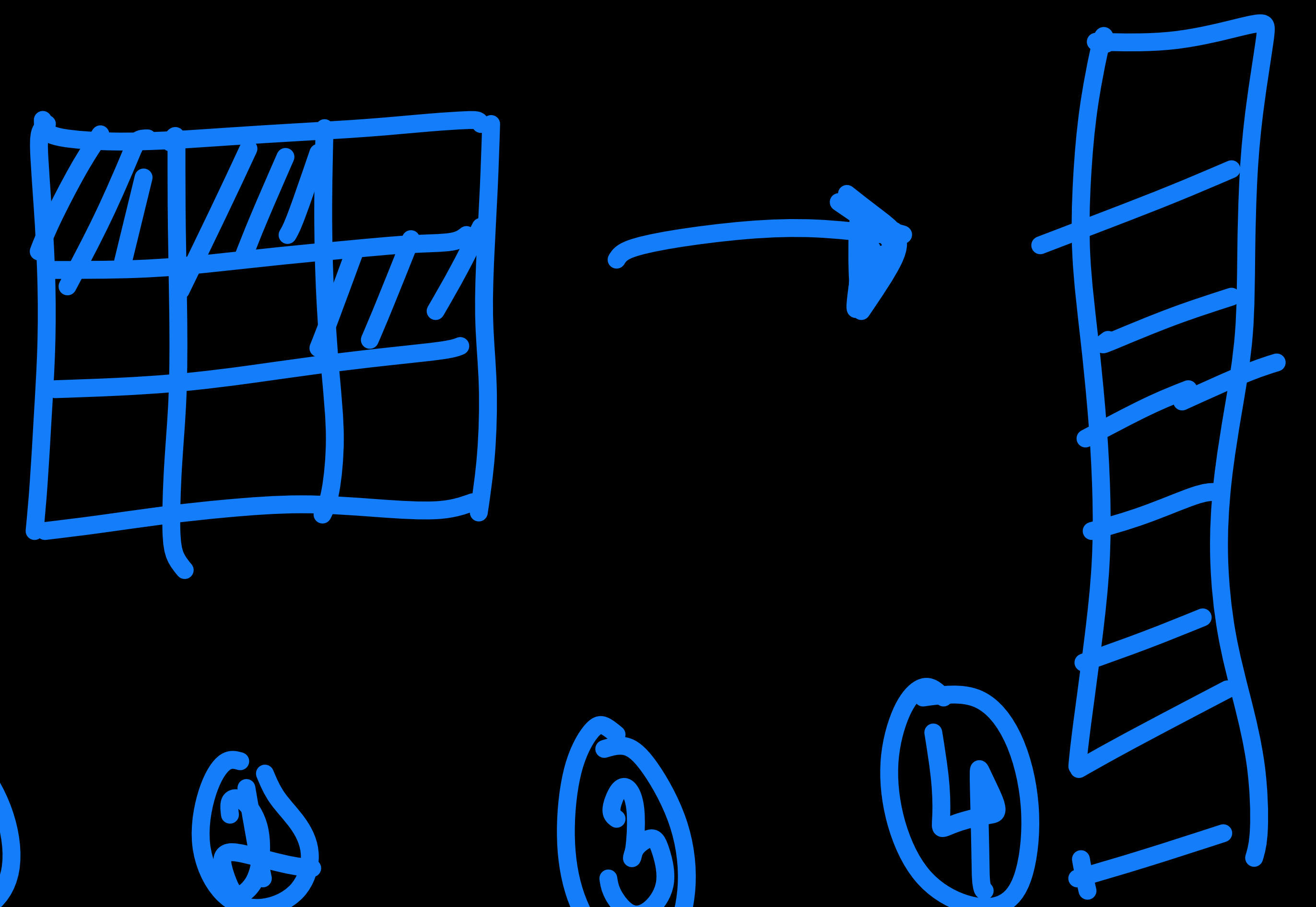
B = % of total # connections

- Oracle pruning
  - Identify each parameters and its relevance to the output score
  - Arrange them in increasing order of relevance
  - Preserve only the most relevant.
  - Computationally complex

Pruning for neural network analysis

- Minimum weight criteria

$$\frac{1}{|\mathbf{w}|}\sum_{i=1}^{9}|w_i|^2$$



① ② ③ ④

128 128 256 256 . . . .

768

500

- Output activation based
  - ✓ Mean or standard deviation of the output activation
  - ✓ Average percentage of zeros
    - Useful in networks with ReLU activations

percentage of neural activations that are propagated

Pruning based analysis of neural networks

$$\mathbf{h} = \{\mathbf{z}_1^{(1)}, \mathbf{z}_2^{(1)}, \ldots, \mathbf{z}_{C_L}^{(L)}\}$$

h = { 1 → $z_1 \cdots z_{C_1}$; L → $z_1^L \cdots z_{C_L}^L$ }

- Taylor series expansion based

$$E(\mathcal{D}|\mathbf{h}_i) \approx E(\mathcal{D}|\mathbf{h}_i = 0) + \frac{\partial E}{\partial \mathbf{h}_i}\mathbf{h}_i$$

- Criterion for pruning feature maps

$$c_i = \left| \frac{\partial E}{\partial \mathbf{h}_i}^T \mathbf{h}_i \right|$$

Pruning based analysis of neural networks

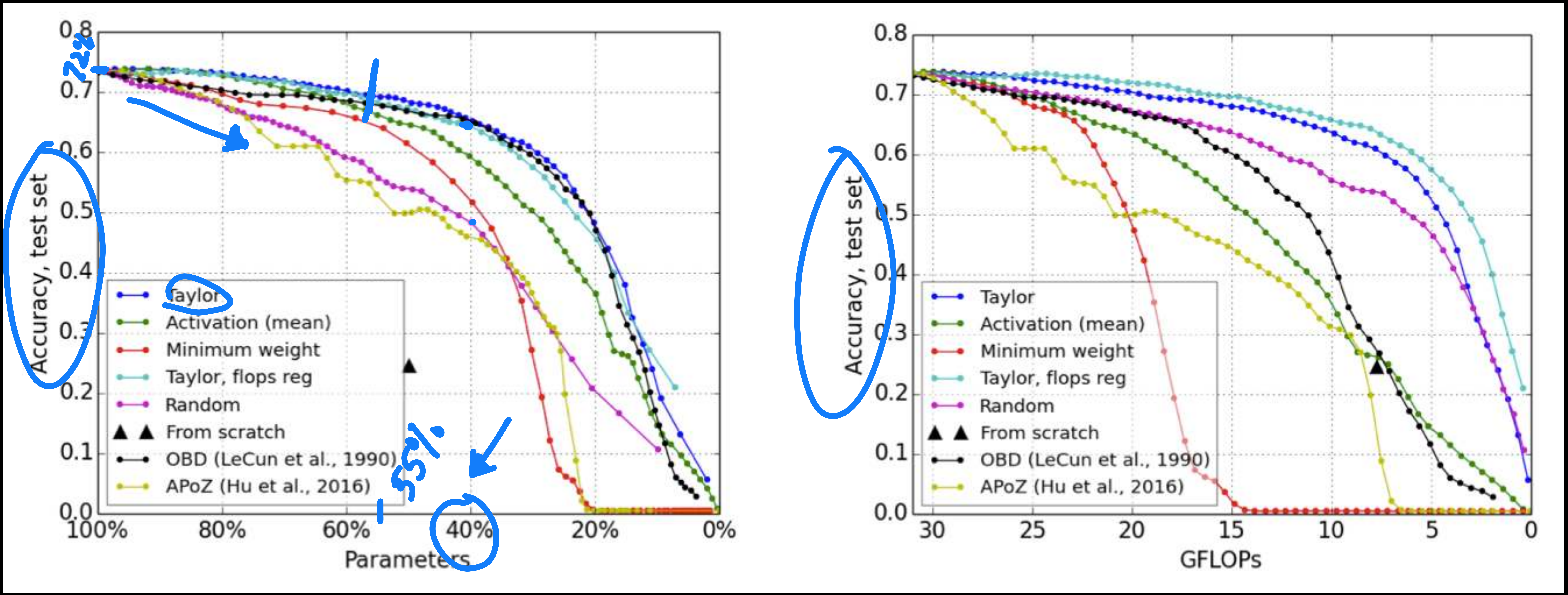
5%
10%
15%

- Correlation with oracle pruning

Corr. of outputs with oracle pruning

	AlexNet / Flowers-102						VGG-16 / Birds-200					
	Weight	Activation			OBD	Taylor	Weight	Activation			OBD	Taylor
		Mean	*S.d.*	*APoZ*				*Mean*	*S.d.*	*APoZ*		
Per layer	0.17	0.65	0.67	0.54	0.64	**0.77**	0.27	0.56	0.57	0.35	0.59	**0.73**
All layers	0.28	0.51	0.53	0.41	0.68	0.37	0.34	0.35	0.30	0.43	0.65	0.14
(w/ ℓ_2-norm)	0.13	0.63	0.61	0.60	-	**0.75**	0.33	0.64	0.66	0.51	-	**0.73**

Pruning based analysis of neural networks



Adversarial Examples and Explainability

Adversarial examples and learning

Adversarial Examples: Attacks and Defenses for Deep Learning

Xiaoyong Yuan, Pan He, Qile Zhu, Xiaolin Li*
National Science Foundation Center for Big Learning, University of Florida
{chbrian, pan.he, valder}@ufl.edu, andyli@ece.ufl.edu

Adversarial attacks and model defenses

- Examples that fool the model
  - Using a trained image classifier published by a third party, a user inputs one image to get the prediction of class label. Adversarial images are original clean images with small perturbations, often barely recognizable by humans. However, such perturbations misguide the image classifier

$$\begin{aligned} \min_{x'} \quad & \|x' - x\| \\ s.t. \quad & f(x') = l', \\ & f(x) = l, \\ & l \neq l', \end{aligned}$$

Types of Adversarial attacks

- **False positive attacks** generate a negative sample which is misclassified as a positive one (Type I Error). In a malware detection task, a benign software being classified as malware is a false positive. In an image classification task, a false positive can be an adversarial image unrecognizable to human, while deep neural networks predict it to a class with a high confidence score.

- **False negative attacks** generate a positive sample which is misclassified as a negative one (Type II Error). In a malware detection task, a false negative can be the condition that a malware (usually considered as positive) cannot be identified by the trained model. False negative attack is also called machine learning evasion. This error is shown in most adversarial images, where human can recognize the image, but the neural networks cannot identify it.

Types of Adversarial attacks

- White box versus Black box

- Targeted versus Non-targeted

- One-time versus many time

Simple adversarial attack

- Fast Gradient Sign Method

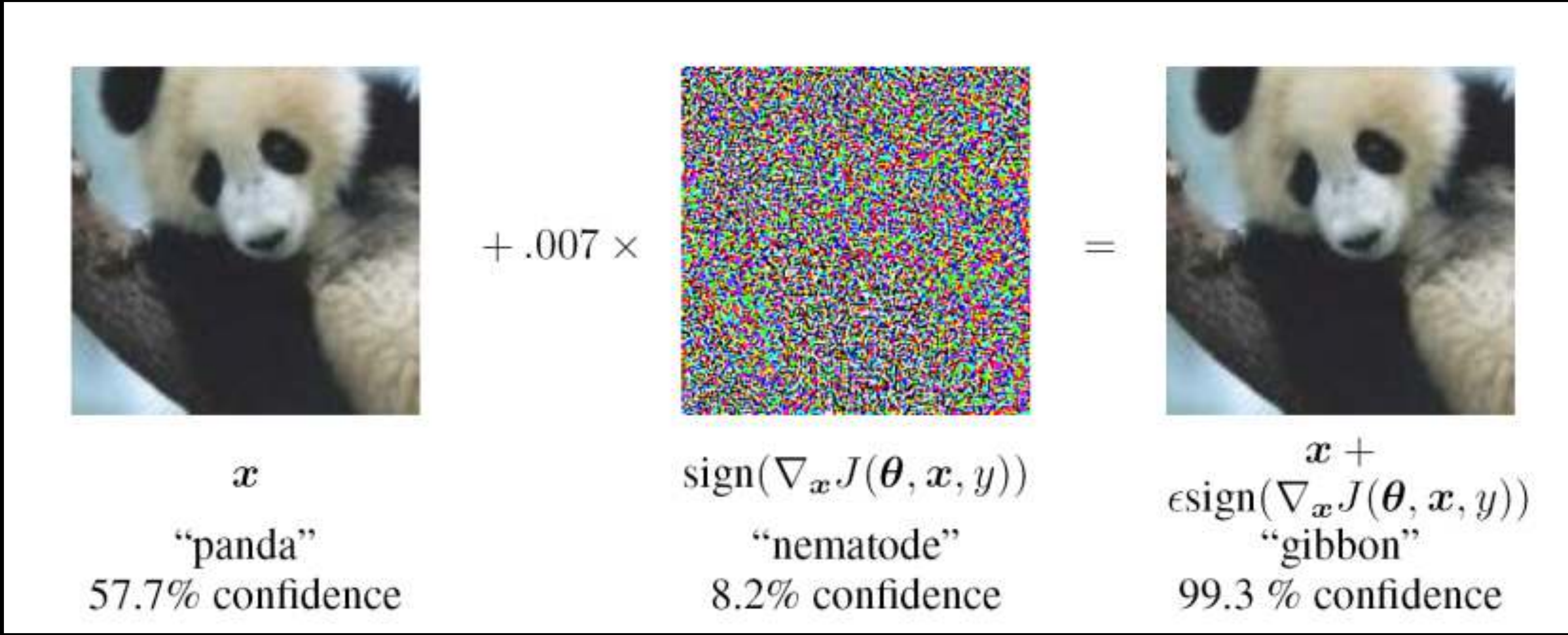
$$\mathbf{x}_i = \mathbf{x}_i + \epsilon sign(\frac{\partial E(\mathbf{x}_i, l)}{\partial \mathbf{x}_i})$$

- Move in the direction of the gradient ascent
- Other similar rules - gradient value based adverasarial learning

$$\mathbf{x}_i = \mathbf{x}_i + \epsilon(\frac{\partial E(\mathbf{x}_i, l)}{\partial \mathbf{x}_i})$$

Simple adversarial attack

- Fast Gradient Sign Method



Adversarial examples

Adversarial Examples for Evaluating Reading Comprehension Systems

Robin Jia
Computer Science Department
Stanford University
robinjia@cs.stanford.edu

Percy Liang
Computer Science Department
Stanford University
pliang@cs.stanford.edu

Adversarial examples in text

- Using similar methods to gradient based update.
- Adding sentences confuses models which will typically not confuse humans

Article: **Nikola Tesla**
Paragraph: "*In January 1880, two of Tesla's uncles put together enough money to help him leave Gospić for Prague where he was to study. Unfortunately, he arrived too late to enroll at Charles-Ferdinand University; he never studied Greek, a required subject; and he was illiterate in Czech, another required subject. Tesla did, however, attend lectures at the university, although, as an auditor, he did not receive grades for the courses.*"
Question: "*What city did Tesla move to in 1880?*"
Answer: *Prague*
Model Predicts: *Prague*

AddAny

Randomly initialize *d* words:

spring attention income getting reached

↓ Greedily change one word

spring attention income other reached

↓ Repeat many times

Adversary Adds: ***tesla move move other george***
Model Predicts: *george*

Adversarial attacks

- ✴️ Understanding adversarial attacks
  - ✓ Allows explainability
  - ✓ Build defenses to these attacks